Morphism and Meaning

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1 Introduction

Herein, we discuss the notions of morphism, interpretation, and meaning by examining the formal systems of sets, numbers, and the *pq-system* described in *Godel, Escher, Bach*[Hof00].

We argue that the meaningfulness of numbers is not inherent, but rather acquired as interpretations of mathematical models of more concrete phenomena. We discuss such models and corresponding interpretations.

1.1 Background

For typographical brevity, let $\mathbb{Z} = \{\cdots, -3, -2, -1, 0, 1, \dots\}$ denote the set of integers and $\mathbb{Z}_+ = \{1, 2, 3, \dots\} \subseteq \mathbb{Z}$ denote the subset of positive integer elements, $\mathcal{S}et$ denote the class of (small) sets, and $\mathcal{F}in \subseteq \mathcal{S}et$ denote the subclass of those sets which are finite. We expect the reader to have some familiarity with *ZF (Zermelo-Fraenkel) set theory*, and understand that $\mathbb{Z}_+ = \langle \mathbb{Z}_+, +, < \rangle$ is a totally ordered, commutative semigroup, that is a set $\mathbb{Z}_+$ of elements with a binary operation $+$ which is commutative and associative and an ordering $<$ which is anti-symmetric, transitive, and connex.

2 Numbers

The *pq-system* is an infinite set of axioms for manipulating the set of symbols $\{p, q, -\}$ according to an axiom schema, a small set of rules, which appear foreign and meaningless. The author declares it to be meaningful because of the following isomorphic interpretation between it and $\mathbb{Z}_+$:

$$\begin{aligned} p &\iff + \\ q &\iff = \\ - &\iff 1 \\ -- &\iff 2 \\ --- &\iff 3 \\ &\text{etc.} \end{aligned} \tag{1}$$

This tacitly assumes that the latter is meaningful to the reader.

I argue by example that interpretation is more adequately expressed by a one-way correspondence, and that the assignation of meaningfulness depends on prior experience.

A reader experienced in arithmetic now learning the *pq-system* may find it to be meaningful by relating it to arithmetic according to the right-ward map $(\rightarrow)$

$$\begin{array}{rcl} p \rightarrow + & & p \leftarrow + \\ q \rightarrow = & & q \leftarrow = \\ - \rightarrow 1 & (\rightarrow) \quad (\leftarrow) & - \leftarrow 1 \\ -- \rightarrow 2 & & -- \leftarrow 2 \\ --- \rightarrow 3 & & --- \leftarrow 3 \\ \text{etc} & & \text{etc}, \end{array}$$

while a reader without experience in arithmetic may find the left-ward map $(\leftarrow)$ to be more meaningful if she had prior experience with the *pq-system*.

Definition 2.1 (morphism: syntax $\rightarrow$ semantics)**.** Let A and B be formal systems of stuff with structure satisfying certain properties. A *morphism*[1] $f : A \rightarrow B$ is precisely an assignment of stuff and structure from the *syntax* A to the *semantics* B which coherently preserves all the necessary properties.

In this case, both maps are bijective morphisms of semigroups of stuff (elements) with structure (order, addition) and properties (totality, associativity, commutativity, etc). Thus, they communicate information according to the definition above in a lossless way according to the definition below.

Definition 2.2 (bijectivity = injectivity + surjectivity)**.** A morphism $f : A \rightarrow B$ is *injective* if it maps distinct stuff in A to distinct stuff in B, and it is *surjective* if all the stuff in B is assigned to some stuff in A. It is *bijective* iff it is both injective and surjective.

The maps are actually isomorphisms because they allow back-and-forth communication in a seamless way.

Definition 2.3 (isomorphism)**.** A morphism $f : A \rightarrow B$ is an *isomorphism* if it is bijective and there is an inverse morphism $f^{-1} : B \rightarrow A$ such that $f \circ f^{-1}$ and $f^{-1} \circ f$ don't do anything.

Thus, the fundamental concepts here are one-way morphisms and isomorphisms arise only when both one-way morphisms exist and cooperate in a sensible fashion which does not always occur. For example, there is a non-surjective

[1] In many areas of mathematics, the word "homomorphism" is used instead to emphasize the "homo-", that is homogeneity or sameness, underlying the algebraic structures. One can properly define the concept of "morphism" in any well-defined theory, although they are usually taken as axiomatic gadgets in category theory.

way to interpret the *pq-system* using the natural numbers $\mathbb{N} = \mathbb{Z}_+ \cup \{0\}$ under addition $+$, but there is no interpretation of $\mathbb{N}$ using the *pq-system* since there is no part corresponding to 0.

Of course, it is possible to be fluent in both systems and yet attach to meaning to either; these interpretations merely serve to transfer meaning if it already exists. Someone learning both systems simultaneously may identify a bijective correspondence without attaching meaning to either.

In fact, I believe arithmetic (and other formal systems) only have meaning to us because of our prior experience. Maybe it's meaningful to you because it serves as an interpretation for financial transactions, but it wouldn't be meaningful for that reason to a child (or a communist) who does not see the value or the meaning of money. There is a sense in which money is meaningful because it emerges naturally from any sufficiently sophisticated bartering system[Vaz09].

However, I don't think numbers are meaningful in their own right for their own sake. We learn early on that they are merely abstract formalisms which interpret more concrete concepts like cardinality, magnitude, and length. That is, we learn numbers to have a way of talking about a set of cookies (cardinality) we'd like to consume or to argue that your sister's bedroom is unfairly larger than yours (area by square footage).

In the next section, we prove that numbers can be interpreted by a more fundamental concept.

3 Sets

First, we ask that the reader recall the arithmetic system for manipulating sentences involving the natural numbers $\mathbb{N} = \{0, 1, 2, \ldots, n-1, n, n+1, \ldots, \}$; such a system can be formalized using the *Peano axioms*.

We interpret the natural numbers with finite sets by constructing a bijective morphism $S : \mathbb{N} \to \Sigma$, that is a way of assigning to $\cdot$ in $\mathbb{N}$

$$
\begin{aligned}
< &\to S(<) = \cdots \\
+1 &\to S(+1) = \cdots \\
+ &\to S(+) = \cdots \\
\times &\to S(\times) = \cdots \\
\div &\to S(\div) = \cdots \\
\hat{} &\to S(\hat{}) = \cdots \\
0 &\to S(0) = \cdots \\
1 &\to S(1) = \cdots \\
2 &\to S(2) = \cdots \\
n &\to S(n) = \cdots \\
&\text{etc}
\end{aligned}
$$

something $S(\cdot)$ in $\Sigma \subseteq \mathcal{F}in$ which captures the essence of the stuff (elements),

structure (order, succession, addition, multiplication, division, and exponentiation), and properties (associativity and other relations where applicable) in a way that is coherent with the rules, axioms, and theorems.

We begin by constructing the formal system of *von Neumann ordinals* which is a an infinite collection Σ of finite sets and an infinite set of axioms, rules, and theorems about containment therein.

In doing so, we implictly rely on the *ZF axioms of extensionality, regularity, specification, pairing, union, replacement, infinity, power, and well-ordering*, the third of which proves the existence of an empty set, denoted $\emptyset = \{\}$.

We include as axioms that $\emptyset \in \Sigma$ and that

$$X \in \Sigma \Rightarrow X \cup \{X\} \in \Sigma.$$

This lets us define

$$\begin{aligned} 0 &\mapsto S(0) = \emptyset \\ 1 &\mapsto S(1) = \{\emptyset\} \\ 2 &\mapsto S(2) = \{\emptyset, \{\emptyset\}\} \\ 3 &\mapsto S(3) = \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\} \\ n+1 &\mapsto S(n+1) = S(n) \cup \{S(n)\} \\ &\text{etc,} \end{aligned}$$

which suffices as a morphism of infinitely generated sets.

To progress, we require the inverse morphism $|\cdot| : \mathcal{F}in \to \mathbb{N}$ which assigns to a set its cardinality

$$\begin{aligned} 0 &\hookleftarrow \emptyset \\ 1 &\hookleftarrow \{\emptyset\} \\ 2 &\hookleftarrow \{\emptyset, \{\emptyset\}\} \\ 3 &\hookleftarrow \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\} \\ &\text{etc,} \end{aligned}$$

which holds for all von Neuman ordinals as well as arbitrary finite sets, e.g. $|\{a, b, \{\text{apple}\}, 6\}| = 4$. This allows us to define the morphism $\bar{\cdot} : \mathcal{F}in \to \Sigma$ as the composition $\bar{\cdot} = S \circ |\cdot|$ which gives the von Neuman ordinal corresponding to its cardinality. Thus, Σ inherits structure from $\mathcal{F}in$ by applying its operations and converting via $\bar{\cdot}$. In particular, there is a cardinality morphism $|\cdot| : \Sigma \to \mathbb{N}$ on Σ defined by restricting that of $\mathcal{F}in$.

Let $n, m \in N$. Note that the we can generate n by applying the successor relation once to $S(0)$ to obtain $S(1) = S(0{+}1)$, twice to obtain $S(2) = S(1{+}1) = S(0+1+1)$, and n times to obtain $S(n) = S(n-1+1) = S(0+1+1+\cdots+1)$. Any element $X \in \Sigma$ is the realization of applying the successor relation $|X|$ times to $S(0)$, so the morphism is surjective. If $n < m$, then $S(n) \subset S(m)$ and thus $|S(n)| < S(m)$. This also proves that the morphism is injective since distinct numbers yield distinct ordinals of distinct cardinalities.

We define $S(n)+S(m) = \overline{S(n) \sqcup S(m)}$ to be the von Neuman ordinals whose cardinality is the that of the disjoint union of $S(n) \sqcup S(m)$. We also define $S(n) \times S(m) = \overline{S(n) \times S(m)}$ to be the von Neuman ordinal whose cardinality is that of the usual Cartesian products of sets. If $n/m \in \mathbb{N}$, then we can define $\overline{S(n)/S(m)}$ to be the ordinal whose cardinality of some set obtained by taking the quotient set of $S(n)$ according to some equivalence relation defined by $S(m)$ which always exists as needed according the well-ordering theorem. Finally, we define $S(n)^{S(m)} = \mathrm{H}\,om(S(n), S(m))$ to be the ordinal whose cardinality is the number of functions from $S(n)$ to $S(m)$.

We have now assigned all of the stuff and structure we endowed on the natural numbers $\mathbb{N}$, and proven that the mapping is bijective. It remains to show that all of the properties are satisifed. We shall not do so here, and instead encourage the reader to do so. This would finish the proof that there is a a bijective isomorphic interpretation between the natural numbers and the von Neuman ordinals.

We can also expand this to be a morphism $S^{-1} : \Sigma \to \mathbb{C}$ which offers meaning regarding subtle phenomena about negative, irrational, transcendental, and imaginary numbers that does not apply to von Neumann ordinals.

4 Reasoning

We can expand S to $S : \mathbb{N} \to \mathcal{F}in$ which fails to be surjective. I argue that this is the intuitive interpretation of natural numbers; this correspondence highlights that numbers are abstractions that model sizes of collections. We could instead envision a different morphism from real numbers to four-dimensional spacetime which highlights that length and proper time are abstractions by interpreting them as certain features of physical spacetime.

That is, the Peano axioms do not endow numbers with meaning a priori. Instead, numbers acquire meaning from the existence of morphic interpretations to other formal systems (e.g. economic utility, finite sets, spacetime) that model something which we derive meaning from via experiences derived from "direct" perceptual observation.

References

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